Cryptography FYP Projects

AI, blockchain, crypto, and decentralization.
Build on real results.
Create things you'll be proud of.

Sherman S. M. Chow

smchow, 808

1 Private AI / verifiable ML

2 Private fuzzy matching

3 Encrypted search

4 Threshold wallets

5 Decentralized credential

6 Cross-chain exchange

7 AML / CBDC payments

8 AI security / safety

9 Systems security

Self-Motivation is Crucial

Artifact-backed cryptosystems -> Blockchain/privacy applications -> Broader security families

PROJECT 1

Privacy-Preserving AI and Verifiable Machine Learning (ML)

Run an ML model on private data without revealing the input, or prove that a large ML-style computation was done correctly without asking others to rerun it.

FOUNDATION

- Start from a system for private transformer inference (SHAFT)
- or a framework for proof-backed matrix computation for verifiable deep learning (Evalyn).

WHAT YOU DO

- Begin with one real artifact and push it one step further:
- a new model, a new workload, a new deployment setting,
- or a stronger implementation.

PAYOFF

- A working cryptographic ML system that you can run, modify, and explain,
- built directly on top of a real research artifact.

Private Fuzzy Matching

Real matching is noisy.
Match records that are similar but not identical
without showing either side's records to the other side.

FOUNDATION

- Private approximate matching:
- cryptographic techniques that let two sides detect "close enough" records
- without revealing the records themselves.

WHAT YOU DO

- Turn that capability into one realistic workflow,
- such as name screening, watchlist matching, or
- noisy record linkage between two databases.

PAYOFF

- A private matching system that still finds the suspicious or relevant near matches,
- instead of missing them just because the strings are not exactly equal.

PROJECT 3

Multi-Client Searchable Encryption over MongoDB

**Fast search over encrypted data is already hard.
Doing it for many writers and many readers
while keeping search sublinear was long viewed as out of reach.**

FOUNDATION

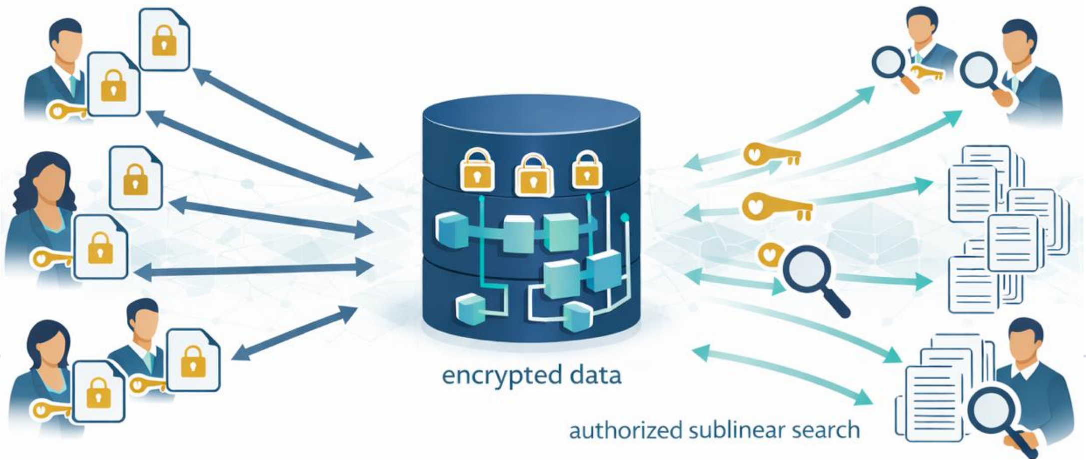
- Start from recent work that cracks open a long-standing open problem —
- sublinear multi-client searchable encryption

WHAT YOU DO

- Build a MongoDB-backed prototype for multi-user encrypted search

PAYOFF

- A real backend + frontend: many users contribute data, authorized users can search,
- and the server never gets plain access to the documents.



PROJECT 4

Threshold Wallets and Robust Shared Control

Authorize a blockchain transaction
without giving any one person the full signing key,
and keep the wallet usable even when one signer is missing, delayed, or misbehaving.

FOUNDATION

- Start from recent work that makes true threshold signature survive real faults and aborts,
- and follow-up work that cuts the cost of shared signing.

WHAT YOU DO

- Build toward a real wallet flow:
- shared approval, policy checks, signer replacement, recovery,
- and fault handling under realistic failure cases.

PAYOFF

- A wallet that removes the single-key bottleneck
- without turning every transaction into a coordination nightmare.

PROJECT 5

Anonymous Credential Applications on Decentralized Platform

Prove that you are trustworthy / eligible / reputable enough without revealing who you are or your full activity history.

FOUNDATION

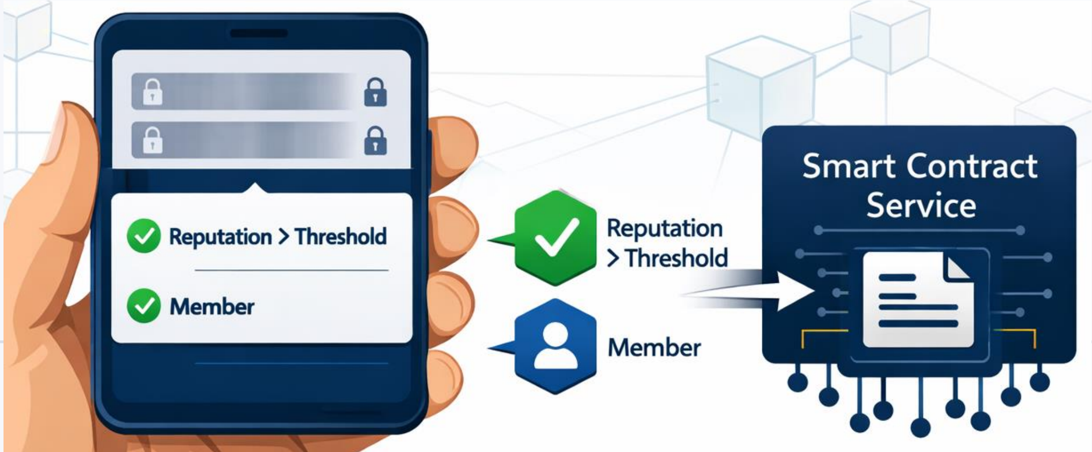
- Start from scored anonymous credentials with reputation and revocation mechanisms,
- or anonymous transaction that hide linkage while detecting double-spending

WHAT YOU DO

- Build anonymous credential or anonymous transaction
- into a usable decentralized / blockchain application.

PAYOFF

- A user interacting with a blockchain application in a privacy-preserving way:
- the system accepts actions but does not learn about who the user is or their full history



PROJECT 6

Privacy-Preserving Atomic Cross-Chain Exchange

Swap assets across chains
so that either both sides complete the trade or neither does,
while exposing the full deal to everyone watching the chain.

FOUNDATION

- Start from recent work on privacy-preserving atomic cross-chain exchange,
- where atomicity and privacy are designed together

WHAT YOU DO

- Build the application flow:
- privately propose a swap,
- accept it,
- and complete it atomically across chains.

PAYOFF

- A working prototype where two wallets on different chains complete a swap
- with user experience and privacy protection and user experience as first-class goals.

PROJECT 7

Privacy-Preserving AML and CBDC Payments

Let a payment pass a bounded compliance check without exposing the whole payment record.

FOUNDATION

- Start from privacy-aware payment-compliance design,
- including Project Mandala and current CBDC payment ideas.

WHAT YOU DO

- Build AML-aware payment flow
- that enforces sanctions checks, amount caps, or KYC-tier rules
- with minimal disclosure.

PAYOFF

- A wallet or payment flow where a transfer can pass a sanction screening
- without exposing the whole transaction record to others

PROJECT 8

AI Security and Safety

Keep an AI application useful while making it meaningfully harder to trick into doing the wrong thing.

FOUNDATION

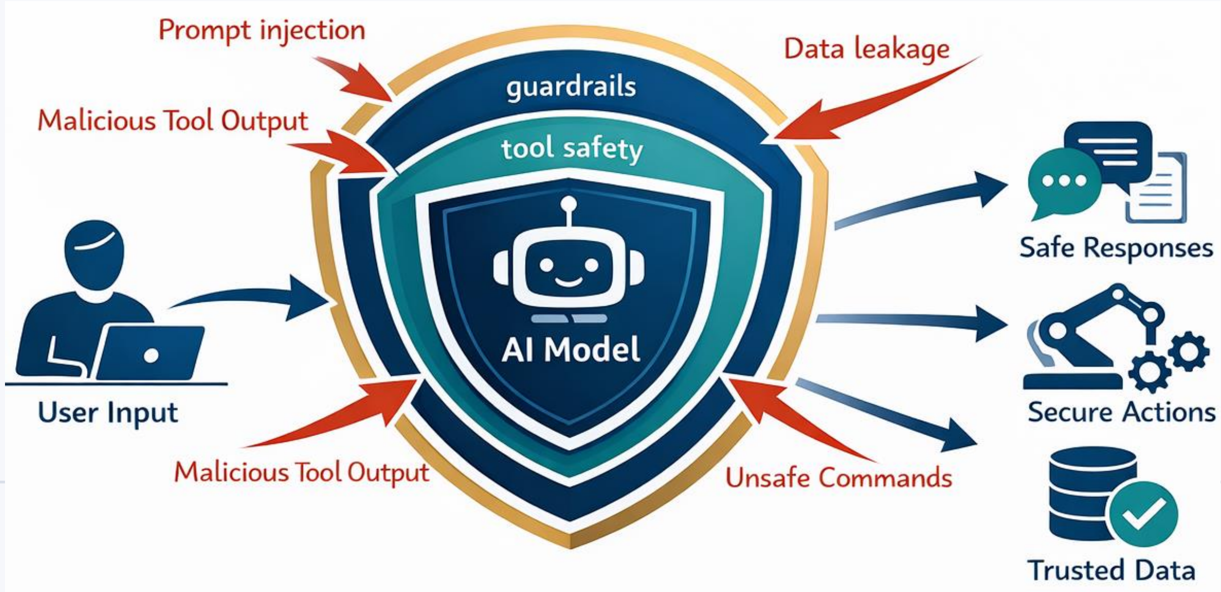
- Start from one concrete attack prompt injection, model misuse, unsafe tool use, or output abuse

WHAT YOU DO

- Harden in one direction —
- a retrieval assistant, an agent,
- a code assistant,
- or a model-based tool

PAYOFF

- An AI system that still works,
- but with a concrete threat model
- and a visible defense story.



PROJECT 9

Systems Security

Take a real system that people already use and make one concrete class of attack harder without rebuilding the whole stack.

FOUNDATION

- Start from a real component and a real weakness:
- authentication, browser, mobile app, sandboxing,
- software supply chain, or service hardening.

WHAT YOU DO

- Build a hardened component, a detector, a policy layer,
- or a reproducible attack-and-defense experiment.

PAYOFF

- A security improvement
- “small” enough for an FYP,
- but “real” enough that you can show off